Helix OTA â€” Emulated / Virtual Device Testing Strategy

Field	Value
Revision	1
Last modified	2026-06-10T00:00:00Z
Status	active â€” Tier-1 in progress
Status summary	The tiered plan for exercising the OTA stack against device-shaped targets without (Tier-1) and with (Tier-2/3) real Android A/B + hardware. Tier boundaries are FACT, established from the hostâ€™s available runtimes and the `containers` submoduleâ€™s capabilities â€” not guesses.
Authority	Helix OTA control-plane / device-integration team
Related	`docs/research/main_specs/CONTINUATION.md`; `docs/RESUMPTION.md`; `containers/` submodule (`vasic-digital/containers`, Â§11.4.76); `submodules/ota-protocol`, `submodules/ota-android-agent`, `submodules/ota-update-engine-bridge`

1. Purpose

The control plane (`server/`), the `ota-protocol` wire contract, and the device-side decision logic (`ota-android-agent` `DeltaApplyDecision`, `ota-update-engine-bridge`) must be validated against device-shaped targets. The realistic targets â€” RK3588 / Orange Pi 5 Max â€” are not always available, and the real Android A/B update path (`update_engine` + AVB/dm-verity + auto-rollback) cannot run on the current Apple-Silicon development host. This doc records the tiered plan that lets each layer be exercised at the highest fidelity the available environment allows, with an honest boundary on what each environment can and cannot prove.

2. Tiers (FACT)

Tier-1 â€” protocol-level device emulator (runnable on THIS macOS host NOW)

- **What:** a podman container running a Go `ota-device-emulator` that speaks the **real** `ota-protocol` to a live control plane and walks the full OTA lifecycle: **register â†’ update-check â†’ telemetry â†’ delta â†’ rollout â†’ recall**.
- **Why it is real (not a mock):** the emulator uses the actual `ota-protocol` types and the actual HTTP `/api/v1` surface â€” it is a real client of the real server, not an in-process fake. Per Â§11.4.27, integration/e2e tiers exercise the real, fully implemented system; the

emulator stands in only for the deviceâ€™s *physical* layer (no real `update_engine`, no real partitions), which Tier-1 explicitly does not claim to cover.
- **What it proves:** wire-contract conformance, server-side flow correctness (anti-downgrade invariant, delta selection + full fallback, rollout cohort advancement, recall = forward-fix), telemetry ingest/read round-trips, multi-device fleet behaviour under concurrent emulated devices.
- **What it does NOT prove (honest boundary):** real A/B slot switch, AVB/dm-verity verification on real partitions, auto-rollback on a corrupt slot, vendor HAL, U-Boot bootloader slot selection. Those are Tier-2/3.
- **Runtime:** podman on this host (the same runtime the pgx integration + dev stack already use via the `containers` submodule). No KVM, no QEMU, no AVD required.
- **Status: in progress** (this is the active PHASE).

Tier-2 â€” Cuttlefish virtual Android device (host/hardware-gated)

- **What:** a Cuttlefish (`cvd`) virtual Android device running a real Android system image, exercising the **real** `update_engine` **A/B flow + AVB/dm-verity + auto-rollback** end-to-end, driven by the `ota-android-agent` + `ota-update-engine-bridge`.
- **What it proves:** the device-side apply path the Tier-1 emulator stubs â€” real payload application to the inactive slot, post-reboot slot promotion, and corrupt-slot â†' auto-rollback.
- **Gate (FACT):** Cuttlefish requires **Linux with nested KVM**. The current development host is Apple-Silicon using the `applehv` hypervisor, which **cannot** run Cuttlefish. Tier-2 therefore requires a Linux CI runner / Linux box with nested KVM.
- **Honest Â§11.4.112 boundary:** Tier-2 is **host/hardware-gated, NOT structurally impossible.** It runs the moment a Linux + nested-KVM environment is available; the blocker is environment provisioning, not a platform/protocol impossibility.
- **Status:** designed, not yet runnable (no Linux/KVM host attached).

Tier-3 â€” real RK3588 / Orange Pi 5 Max hardware

- **What:** the real target board(s) â€” full vendor HAL, real **U-Boot slot-switch**, **dm-verity on real partitions**, real `update_engine`, real auto-rollback on a genuinely corrupt slot.
- **What it proves:** everything Tier-2 proves plus the vendor-specific bootloader and hardware-backed verification that only the physical SoC exercises.
- **Gate (FACT):** requires the physical RK3588 / Orange Pi 5 Max board (operator hardware), per Â§11.4.133 target-hardware-safety discipline for any flash.
- **Honest Â§11.4.112 boundary:** hardware-gated, NOT structurally impossible.
- **Status:** pending hardware availability (the CONTINUATION NEXT-wave items 1â€“2 land here).

3. Tier coverage matrix

Capability	Tier-1 (podman emulator)	Tier-2 (Cuttlefish)	Tier-3 (RK3588)
`ota-protocol` wire conformance	YES	YES	YES
Server flow: register/ update-check/ telemetry/delta/rollout/ recall	YES	YES	YES
	YES	YES	YES

Capability	Tier-1 (podman emulator)	Tier-2 (Cuttlefish)	Tier-3 (RK3588)
Anti-downgrade invariant			
Real `update_engine` A/B apply	no	YES	YES
AVB / dm-verity verification	no	YES (virtual)	YES (real partitions)
Auto-rollback on corrupt slot	no	YES	YES
U-Boot slot-switch / vendor HAL	no	no	YES
Runnable on this macOS host now	**YES**	no (Linux+KVM)	no (hardware)

4. Reuse of the `containers` submodule (Â§11.4.76, extend-donâ€™t-reimplement Â§11.4.74)

The `vasic-digital/containers` submodule (`digital.vasic.containers`) is the canonical containerization machinery and already provides the primitives each tier needs:

- `pkg/boot` + `pkg/compose` + `pkg/health` â€" on-demand boot, compose orchestration, and readiness gating. Tier-1â€™s podman emulator + control plane stack boot via these (the same path the pgx integration test already uses), satisfying the Â§11.4.76 on-demand-infra invariant (the test entry point boots the infra; operators never start podman by hand).
- `pkg/emulator` â€" multi-target Android emulator orchestration (AVD, **x86_64**, with KVM acceleration gating, AVD-lock clearing, orphan `qemu-system-*` reaping). Relevant to an x86_64-AVD variant of Tier-2 on a Linux/KVM host.
- `pkg/vm` â€" QEMU VM orchestration (**aarch64** via `-machine virt` + AAVMF UEFI), the seam for an aarch64 virtual target on a KVM-capable host.

Extension policy: where a tier needs a primitive the submodule lacks (e.g.Â a Cuttlefish `cvd` lifecycle wrapper, or an OTA-specific device-emulator harness), we **extend the submodule upstream via PR** per Â§11.4.74 â€" never duplicate the machinery in-project. The Tier-1 `ota-device-emulator` itself is OTA-domain logic (it speaks `ota-protocol`), so it lives in the project; the boot/health/compose plumbing around it is the submoduleâ€™s job.

5. Anti-bluff posture (Â§11.4 / Â§11.4.27 / Â§11.4.69)

Each tier produces captured evidence under docs/qa/<run-id>/: Tier-1 captures the real request/response transcripts for the full lifecycle against a live server; Tier-2/3 add the on-device apply/rollback evidence. A tier claiming a PASS without exercising the real system at that tierâ€™s fidelity is a Â§11.4 PASS-bluff. The tier boundaries above are stated as FACT precisely so no tier silently claims coverage it does not have.